BMC SWITCHGEAR AND INSTRUMENT TRANSFORMER

BMC
EXCELLENCE REDEFINED™
Since 1986

TECHNOLOGY CENTRE

History is replete with giant companies that rose from humble beginnings. The great Indian engineer, Sir Vishveshwariya pronounced the immortal words- "Industrialize or perish". In 1984, this stirred the minds of two Indian engineers Santanu Mitra & Subodh Banerjee from Kolkata. Later on they embarked on a journey of excellence in the industry along with their team of directors Rajiv Bhargava, Siddhartha Mitra and Girish Magre.

Vision

Our vision is to get recognised as one of the best "Global Medium Voltage R&D House". While carrying out the in-house R&D with "confidentially" as our top most focus, we also like to offer our R&D services worldwide along with the product supply support to the "Switchgear Houses" and the "Electrical Research Institutes".

Mission

In the field of medium voltage Switchgear and Instrument Transformers, we:

- Understand the future market needs in technology & product.
- Carry out continuous innovation in product & process design and application protecting the IPR.
- Support customer with in-house world class manufacturing & testing facility.
- Subsidize the supply of finest quality of components & sub assembly to our customers.

USPs of BMC.....

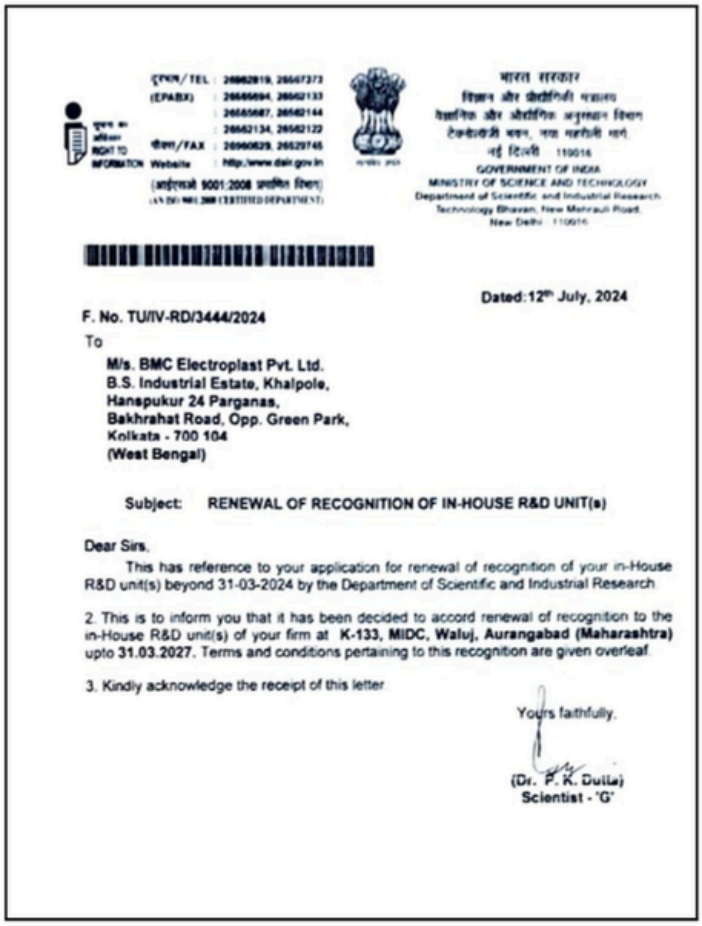
- Patented design of switchgears & mechanisms.
- Pioneers in mechanism design in India.
- Pioneers in Epoxy cast Vacuum Interrupter (Embedded poles)for vacuum contactors worldwide.
- Unique compact design of MV Vacuum contactors.
- Over 10+ patents/design registrations on Switchgears/parts of Switchgears.
- Unique low cost CSS design for windmills in India.
- First in India to offer technology of VCB /VI design to Indian manufacturers.
- First in Asia to offer technology of VCB /VI design to European & overseas companies.
- First in Indian SME sector to get accredited as "R&D centre" in switchgear field by DSIR (India).
- Represent as "Technical/Marketing Consultants" for High Power lab in Korea, Vacuum furnace manufacturer from Germany & component manufacturers for VI (Single window solution on Vacuum Switchgears)
- Experience to design & sell Retrofit VCBs of over 25 versions
- Experience of manufacturing custom built/own designed Epoxy Cast Dry Type Instrument Transformers (CTs/PTs) both for Indoor & Outdoor application
- Experience of manufacturing custom built/own designed Epoxy Cast Switchgear Parts

BMC Certifications

First SME to be accredited as an "R&D" center for medium voltage switchgears by

DSIR (Department of Scientific and Industrial Research) Operating under the Ministry of Science and Technology, Government of India

सत्यमेव जयते
DSIR
DEPARTMENT OF SCIENTIFIC AND INDUSTRIAL RESEARCH



भारत सरकार
विज्ञान और प्रौद्योगिकी मंत्रालय
वैज्ञानिक और औद्योगिक अनुसंधान विभाग
GOVERNMENT OF INDIA
MINISTRY OF SCIENCE AND TECHNOLOGY
Department of Scientific and Industrial Research
Technology Bhavan, New Mehrauli Road,
New Delhi - 110016

Dated: 12th July, 2024

F. No. TU/IV-RD/3444/2024

To

M/s. BMC Electroplast Pvt. Ltd.
B.S. Industrial Estate, Khalpole,
Hanspukur 24 Parganas,
Bakhrahat Road, Opp. Green Park,
Kolkata - 700 104
(West Bengal)

Subject: RENEWAL OF RECOGNITION OF IN-HOUSE R&D UNIT(s)

Dear Sirs,

This has reference to your application for renewal of recognition of your in-House R&D unit(s) beyond 31-03-2024 by the Department of Scientific and Industrial Research.

2. This is to inform you that it has been decided to accord renewal of recognition to the in-House R&D unit(s) of your firm at **K-133, MIDC, Waluj, Aurangabad (Maharashtra)** upto **31.03.2027**. Terms and conditions pertaining to this recognition are given overleaf.

3. Kindly acknowledge the receipt of this letter.

Yours faithfully,

(Dr. P. K. Dutta)
Scientist - 'G'

IMS Certifications of Quality Compliance Management, Environment Compliance Management and Occupational Health and Safety Compliance

QCS MANAGEMENT PVT. LTD.
MANAGEMENT SYSTEMS CERTIFICATION
Certificate of Registration
Awarded to:
ISO 9001:2015 (Quality Management System)
The Quality Management System has been assessed successfully and found to comply with the requirements of the international standard
BMC ELECTROPLAST PVT. LTD.
ADDRESS: 130/22, B.S INDUSTRIAL ESTATE, BAKRAHAT ROAD, KHALPOLE, HANSHPUKUR, OPP GREEN PARK, KOLKATA – 700104, WEST BENGAL, INDIA
Scope of Certification:
DESIGN AND MANUFACTURING OF EPOXY AND NON-EPOXY CAST INSTRUMENT TRANSFORMERS, PRIMARY CURRENT AND VOLTAGE TRANSFORMERS, VCB, VCB RETROFITS AND VACUUM CONTACTORS FOR MEDIUM VOLTAGE APPLICATION AND 11&33 KV INDOOR VCB PANEL.
MANUFACTURE OF EPOXY CAST COMPONENTS FOR ELECTRICAL APPLICATION.
TECHNICAL SERVICE AT CUSTOMER SITE
SUPPLY OF SPARES FOR SUBSTAION EQUIPMENT.
Certificate Number : QCSMPL/QA/0377-R
Initial Issue Date : 23/12/2020
Current Issue Date : 11/12/2025
Expiration Date : 04/12/2026
Re-certification Due Date : 04/12/2026
IRQAO REGISTERED
Partha Bagchi (Managing Director)
QCS MANAGEMENT PVT. LTD.

QCS MANAGEMENT PVT. LTD.
MANAGEMENT SYSTEMS CERTIFICATION
Certificate of Registration
Awarded to:
ISO 14001:2015 (Environmental Management System)
The Environmental Management System has been assessed successfully and found to comply with the requirements of the international standard
BMC ELECTROPLAST PVT. LTD.
ADDRESS: 130/22, B.S INDUSTRIAL ESTATE, BAKRAHAT ROAD, KHALPOLE, HANSHPUKUR, OPP GREEN PARK, KOLKATA – 700104, WEST BENGAL, INDIA
Scope of Certification:
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MANUFACTURE OF EPOXY CAST COMPONENTS FOR ELECTRICAL APPLICATION.
TECHNICAL SERVICE AT CUSTOMER SITE
SUPPLY OF SPARES FOR SUBSTAION EQUIPMENT.
Certificate Number : QCSMPL/QE/0378-R
Initial Issue Date : 23/12/2020
Current Issue Date : 11/12/2025
Expiration Date : 04/12/2026
Re-certification Due Date : 04/12/2026
IRQAO REGISTERED
Partha Bagchi (Managing Director)
QCS MANAGEMENT PVT. LTD.

QCS MANAGEMENT PVT. LTD.
MANAGEMENT SYSTEMS CERTIFICATION
Certificate of Registration
Awarded to:
ISO 45001:2018 (Occupational Health & Safety Management System)
The Occupational Health & Safety Management System has been assessed successfully and found to comply with the requirements of the international standard
BMC ELECTROPLAST PVT. LTD.
ADDRESS: 130/22, B.S INDUSTRIAL ESTATE, BAKRAHAT ROAD, KHALPOLE, HANSHPUKUR, OPP GREEN PARK, KOLKATA – 700104, WEST BENGAL, INDIA
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MANUFACTURE OF EPOXY CAST COMPONENTS FOR ELECTRICAL APPLICATION.
TECHNICAL SERVICE AT CUSTOMER SITE
SUPPLY OF SPARES FOR SUBSTAION EQUIPMENT.
Certificate Number : QCSMPL/QOHS/0379-R
Initial Issue Date : 23/12/2020
Current Issue Date : 11/12/2025
Expiration Date : 04/12/2026
Re-certification Due Date : 04/12/2026
IRQAO REGISTERED
Partha Bagchi (Managing Director)
QCS MANAGEMENT PVT. LTD.

Global Recognition

The Solutions We Can Provide

Comprehensive Digital Monitoring

Covers both temperature and humidity.

Cost-Effective Solutions

Reduces manual monitoring costs

Enhanced Safety

Continuous digital monitoring ensures system integrity.

Originality

All our products are patented & proudly designed and manufactured in India.

User-Friendly Interfaces

Optimal digital controllers with or without displays for versatile monitoring needs.

Long-Term Reliability

Proven technology with installations operational since 1986.

Technology & Manufacturing Expertise

Four Decades of Power System Excellence

OEM Manufacturing Capabilities

Instrument Transformers – CT & PT

Critical Servicing & AMC

Reliable After-Sales & Lifecycle Support

HT Retrofits & Customization

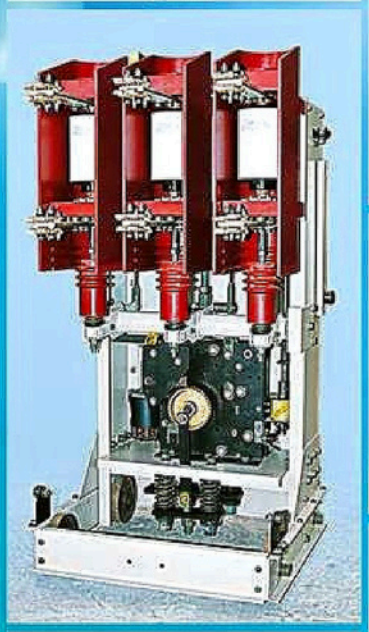
Tailor-Made VBC, VCU, CT/PT & Resin Cast Bushing Retrofit Solutions

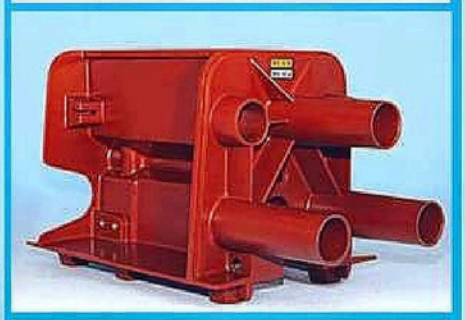
Switchgear Manufacturing Solutions

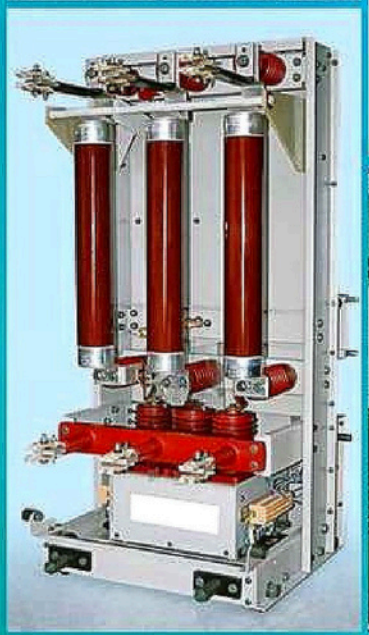
Vacuum Circuit Breakers & Contactors

Cast Resin Component Solutions

Tailor-made Components & Sub-Assemblies







MV Switchgears Components

- Mechanisms for 12, 24 & 36 kV VCBs
- Support insulators
- Drive insulators
- Moulded Spouts upto 5000 Amps
- Epoxy housing three in one
- Epoxy housing single
- Molded Vacuum Interrupters Indoor
- Molded Vacuum Interrupters Outdoor

MV Instrument Transformer

Epoxy Cast

- Indoor CTs upto 44 kA
- Indoor PTs
- Outdoor Dry Type CTs
- Outdoor Dry Type PTs

MV Vacuum Switchgears

- Retrofit Vacuum Circuit Breaker (VCB) Vertical & Horizontal Isolation
- Vacuum Contactors (VC) & Vacuum Contactor Panels (VCP) upto 12kV
- Retrofit with VC & Fuses on a trolley
- Modular Indoor VCB & VC Panel
- Outdoor VCB (Metal Clad & Porcelain Clad)
- VCB with Magnetic Actuator Drive (Indoor and Outdoor)
- Vacuum RMU
- Unitised Substations/Compact Substations
- MV Vacuum Switchgear
- Vacuum Interrupter
- MV Instrument Transformer
- MV Switch Board Panels
- Soft Starters

BORN TO DISCOVER

USPs of CSS

CSS accommodates VCB, CT, PT, CRP, Bus Duct etc. Hence compact in size therefore space required for Panel is less than a half w.rt

- Conventional outdoor arrangements
- Easy & faster in Erection & Commissioning
- Eco-friendly, no Oil based CTs, PTs; no gas filled CBs
- Maintenance Friendly
- Customized Design For Wind Mill Substation
- Minimal Foundation Work
- Low Life Cycle Cost
- Adoptable to various types of Transformers through Bus Ducts
- Safer
- Easy to Transport
- Vandalism Proof
- Typical Reference Standards: IS:14786:2000, IEC 62271-100 & 200 (2012), IS:3156-(Part-1):2007, IS2705(Part:1):2007

BMC's

Compact Sub Station

CSS Accommodates VCB, CT, PT, CRP, Bus Duct

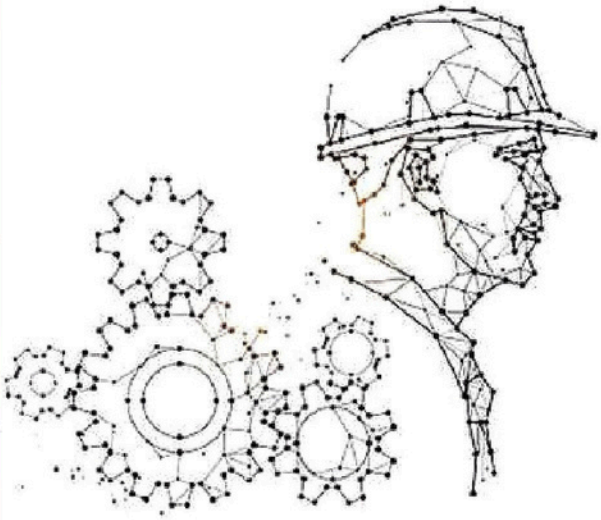
Technical Data		Units	Non-Metering CSS	Metering CSS
Rate Voltage (U_r)		kV	36	36
Rated Insulation Level	Power Frequency Withstand (U_d)	kV 1 min	70	70
	Lightning Impulse Withstand (U_p)	kV peak	170	170
Rated Frequency (f_r)		Hz	50/60	50/60
Rated Normal Current (I_r)		A	800/1600/2000	800/1600/2000
Rated Short-Circuit Breaking Current (I_{sc})		kA	25/40	25/40
Rated Short-Circuit Making Current (I_{ma})		kA peak	62.5/10V	62.5/10V
Rated Duration of Short-Circuit (STC)		Seconds (s)	3	3
CT Ratio	Metering CT	A	–	50/5-5
	Ring Type CT		200/5	–
PT Ratio	Metering PT	V	–	33/√3/110/√3
	Single Ph PT		33kV/rt3/110V	

Note: the Technical Specification & Dimensions can be customised if required.

Our Products

Typical Products for Windmill Applications

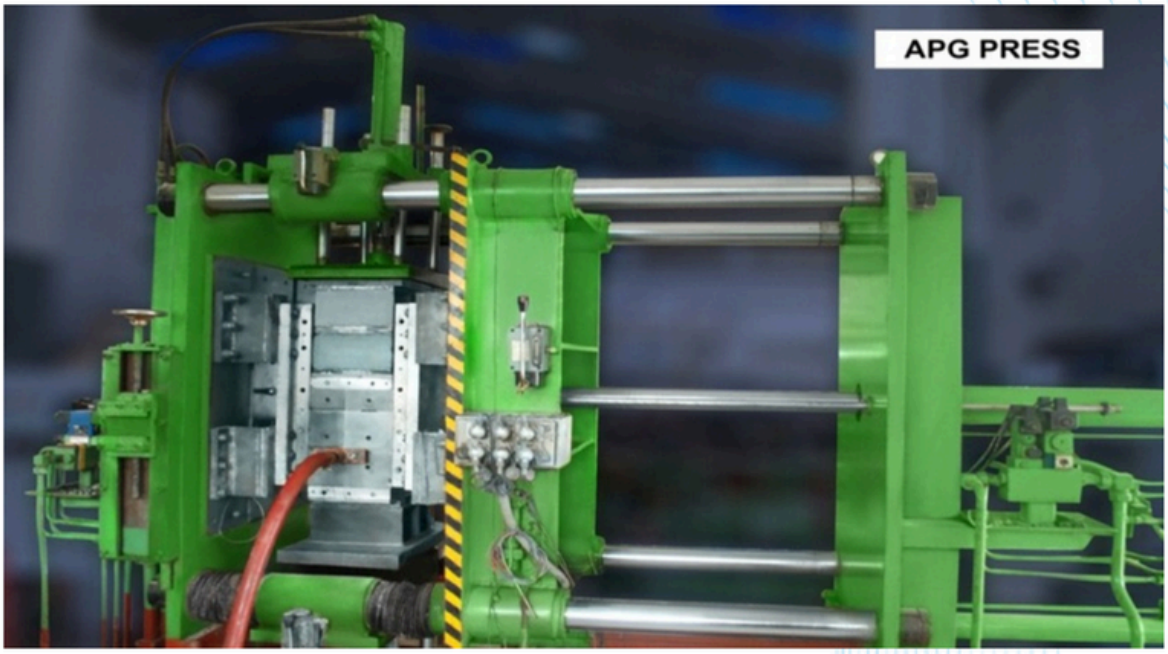
- Compact Substations (CSS) upto 36kV
- Outdoor VCB (Metal Clad & Porcelain Clad) upto 36kV, 40kA
- Control Relay Panel (CRP)
- Vacuum Contactor (VC) & Vacuum Contactor Panels (VCP) upto 12kV
- Indoor Resin Cast CTs upto 12kV
- Indoor Resin Cast PTs
- Outdoor Dry Type Resin Cast CTs
- Outdoor Dry Type Resin Cast PTs
- LT Vacuum Contactor upto 1250A
- Ring Main Unit
- Load Break Switch
- Metering Cubical

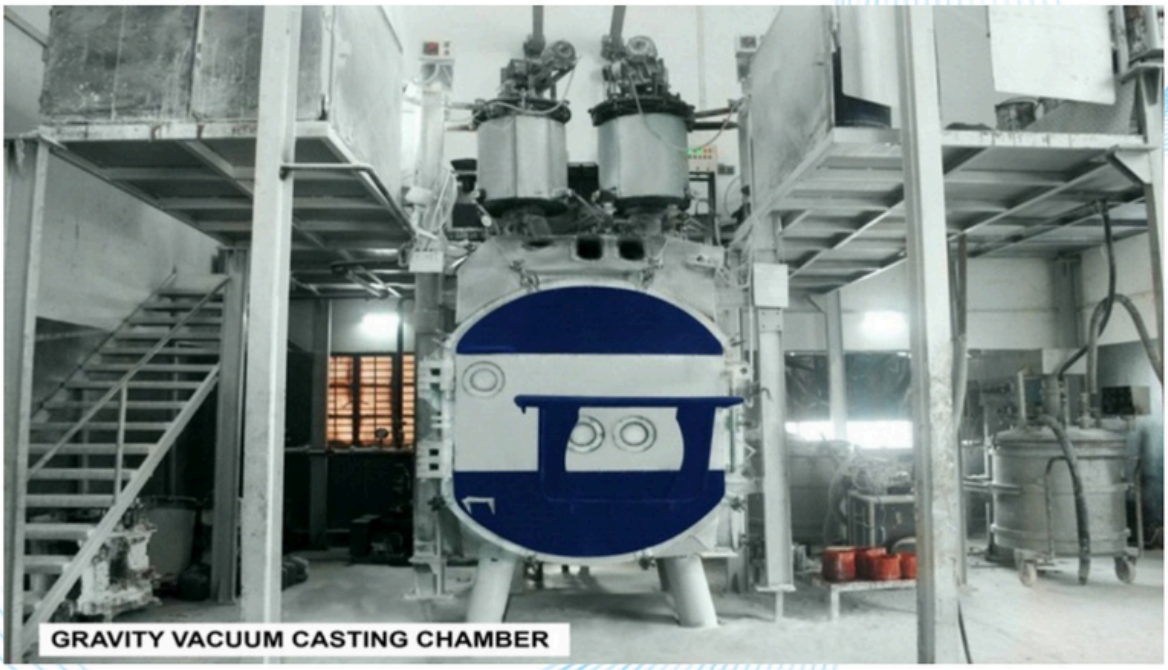


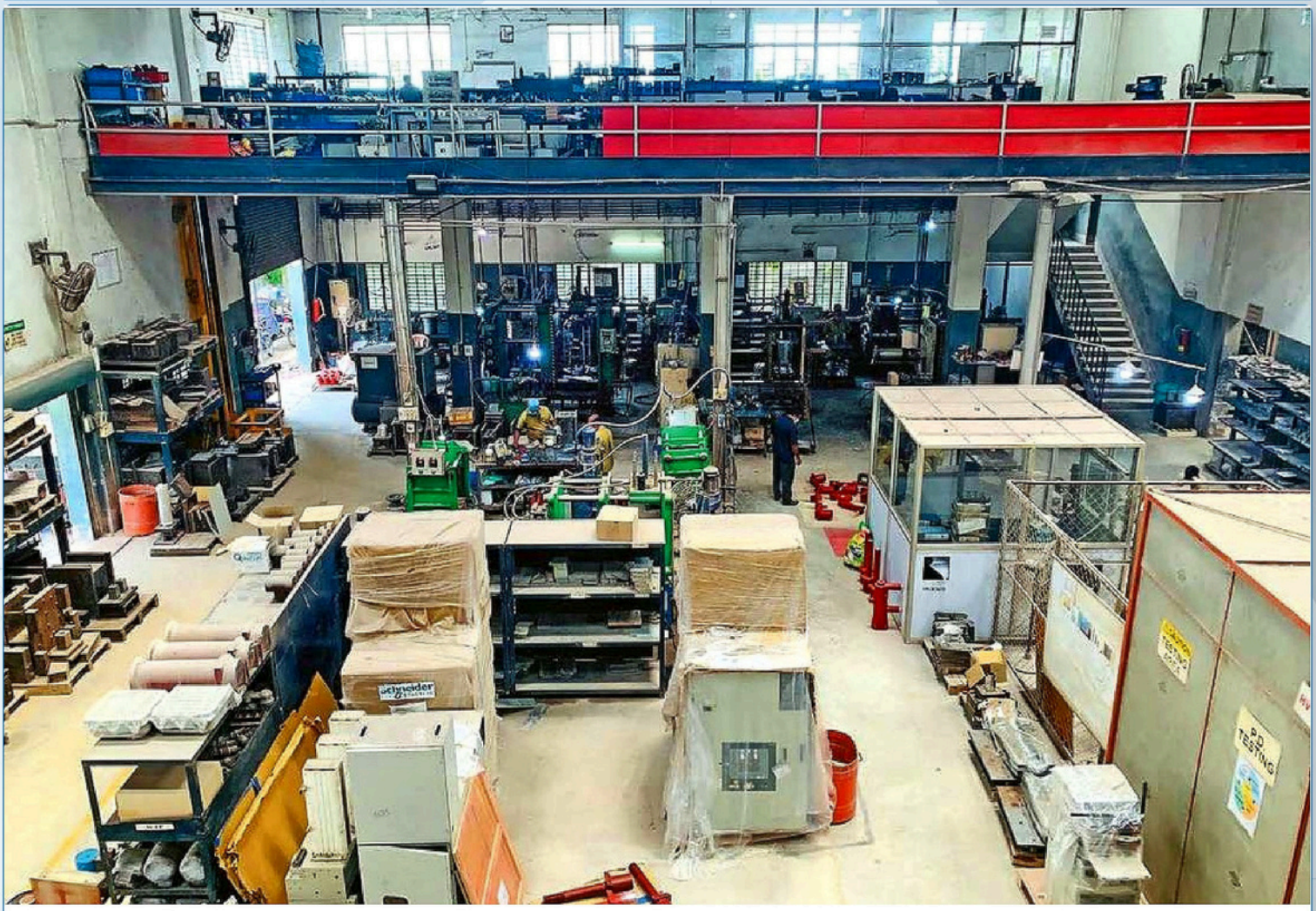
Manufacturing

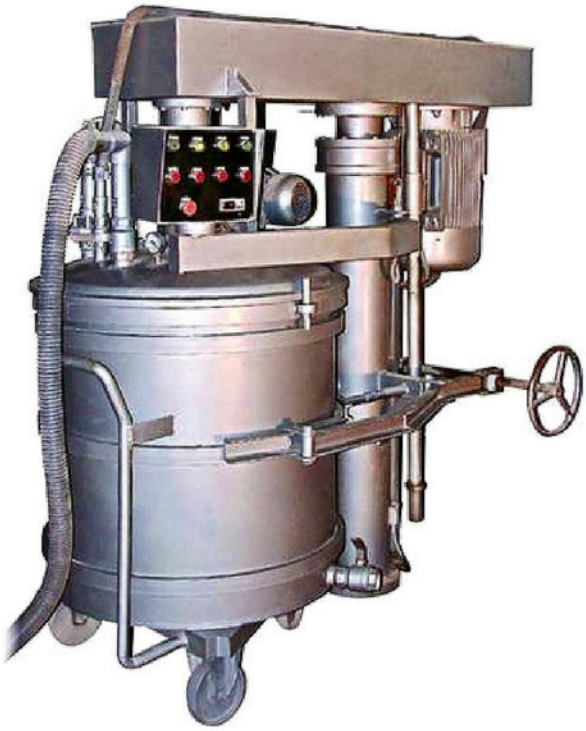
BMC, apart from the design and technology services is equipped with the manufacturing and testing facilities for MV Switchgear Components, MV Switchgear, MV Instrument Transformers, MV & LT Vacuum Contactors These facilities are located at Kolkata & Aurangabad in India.

IN-HOUSE MANUFACTURING FACILITIES :









BMC hub at Kolkata

Manufacturing Facilities:

- Epoxy Casting Plant (Vacuum Casting & APG)
- Coil Winding & Assembly for CTs & PTs
- CRP, VCB & VC Assembly
- Mechanism Assembly
- Plating Shop
- Machine & Forging Shop
- CNC Fabrication (Tie-Up)
- Powder Coating (Tie-Up)
- Automatic Busbar Punching & Bending

APG-our specialty: The Automatic Pressure Gelation technique enables achievement of increased output as well as superior quality.

The advantages are as below-

- Low cycle times.
- Rapid production irrespective of size or complexity of shape.
- Reproducible dimensions within close tolerance.
- Cleaner process with reduced wastages.
- Reduction in rejection percentage.
- Improved mechanical properties and better surface finish

The hydraulic operated presses used make the moulding process accurate, fast, clean and hazard free. The Process Control parameters are accurately maintained with the help of advanced instrumentation and PLC control.

Certified
MV Component
Panel Builder
Schneider Electric

BMC
EXCELLENCE REDEFINED
Since 1986

Manufacturer of MV Switchgear panels as core component advanced partner panel builder with Schneider Electric VCB up to 36kV system voltage.

36 kV SWITCHGEAR PANEL

Advanced Panel Builder And Global Supplier OEM To Schneider Electric For CT and PT

12 kV SWITCHGEAR PANEL

12kV SCHEINDER ELECTRIC VCB

36kV SCHEINDER ELECTRIC VCB

Services Offered By BMC

- DIGITISATION FOR SUSTAINABILITY, HEALTH MONITORING AND DATA ANALYSIS OF COMPLETE POWER SYSTEMS.
- COMPLETE ROUTINE TESTING AND DIAGNOSIS OF SUBSTATION EQUIPMENT AT SITE.
- REPAIR, REPLACEMENT AND RETROFITTING OF COMPLETE SUBSTATION EQUIPMENT.
- TAILOR MADE COMPONENTS FOR IMPORTED, CRITICAL AND OBSOLETE SUBSTATION EQUIPMENT.
- PREDICTIVE MAINTENANCE FOR EARLY FAULT AND FIRE RISK DETECTION
- END-TO-END ASSET LIFECYCLE MANAGEMENT WITH SCADA INTEGRATION

USPs of BMC

With Over Four Decades Of Experience In Power Distribution.

Ready To Move Skilled Engineers And Technicians With Hands-On Knowledge.

Cent Percent Problem Rectification On War Footing Basis.

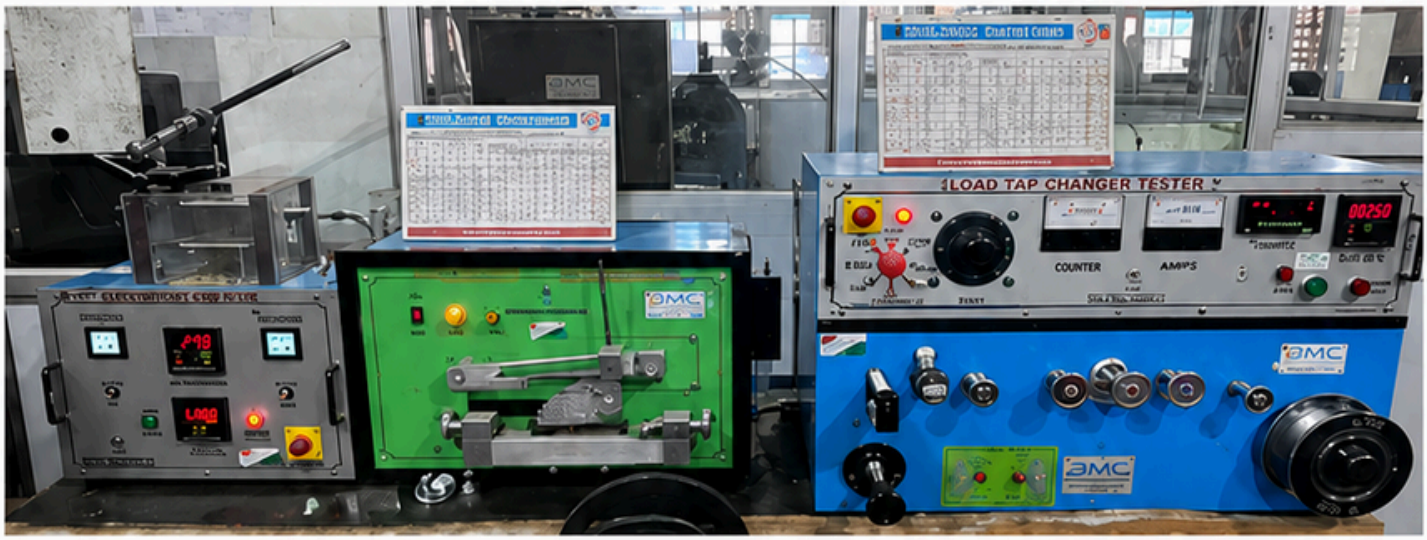
Customized Economic Solution As Per Equipment Health, Installation Site Condition, Operator Suitability And Local Environment Condition.

Solution For Any And All Make Of Equipment From India And Abroad.

Our Strengths

- We are an ISO 9001:2008 certified company.
- The Repetitive works clients have awarded us over the years.
- Our experience in working with most of the leading Switchgear Manufacture like Areva, Schnieders, Crompton Greaves, L&T, BHEL, Easun, Reyrolle, Vijai Electricals, GE, EATON of the Country.
- 100% track record of Completed Projects.
- Our Commitment to Quality, Speed and service.
- A well structured service back up team.
- Our International Presence.

  ...Global Switchgear Players like BATEL (Turkey), Tamco (Malaysia), Johnson, Philips (Dubai), KERI, ABM (Ukraine), Pommier - Group Cahores (France), Ormazabel (Spain) etc.
- Patent registration on many exclusive Product designs viz. Vacuum Interrupters, Vacuum Contactors, Dry type outdoor CTs, Mechanism.
- Type Testing of Product designs in KEMA, KERI, CPRI, ICMET International Labs.



A New Age Research And Design Organization

BMC ELECTROPLAST (I) Pvt. Ltd. The name BMC is synonymous with optimal solution in A non-stop knowledge of electrical products systems, designing of ideas working techniques as as a confidant quality control system, custom no devices of durable and cost effective products and services for every respectable purpose in switchgear. Based to care of the pioneer in saying “No leaps deter our bows such as CFD, Thermal imaging, FEA etc., for designing their product.

Success Not By Accident But By Skill And Sense

BMC ELECTROPLAST is geared to undertake turnkey projects based on a thorough process conceptualization. This process less risks of errors and hazardous to ensure that a client receives products and services of the highest level isa. It real tests, mechanisms provide management to have control over projects and comply minimize chances of project overruns.

Testing Facilities

- CT & PT Test setup
- PD Test setup
- Impulse Testing
- VCB Analyzer
- Mechanism Endurance Test Jig
- HV & MV Drop Test
- Temp rise Test
- Portable Vacuum Checker for Vacuum Interrupters
- Mechanical Testing facilities

Aurangabad the Technology Centre of BMC

A momentous decision was made in 2004, to further expand activities to Aurangabad, an industrial epicentre. Mr. Girish Magre, the enterprising electrical engineer and a global traveller, spearheaded the move. Six years later, the second plant at Aurangabad is designed considering the Eco-friendly green ambience, safe working stations and energy saving building architecture. The new State of Art manufacturing facility at Aurangabad- India has begun to be recognised as a Technology Centre, attracting many overseas Switchgear Houses. Enquiries to align the R&D work with product supply support had started pouring in.

Teamwork, safety, health environment:

At BMC, teamwork is one of the top most agenda. All the projects are discussed regularly at various stages. The active involvement of the team is ensured during the discussions. Though there are work freedom at all levels, the major decisions are taken collectively after principally reaching a consensus. BMC ensures the health & safety of the entire stakeholders.

Our Partners

We are now representing the following companies: BATEL-Turkey, Sirui-China, PVA Tepla-Germany, etc.

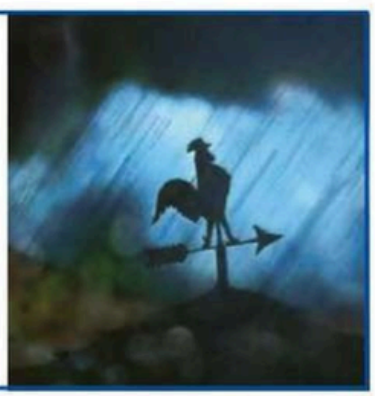
BATEL elektromekanik

Sirui

PVA TePla

A societal responsibility:

We thus ensure contribution to that society as a responsible corporate citizens.



NAMES THAT TRUST US FOR FOUR DECADES

GOVERNMENT ENTITIES

ASSAM STATE TRANSPORT CORPORATION
INDIAN RAILWAYS
IndianOil
CLW
CPWD
बेस्ट
Coal India
भारत इलेक्ट्रॉनिक्स BHARAT ELECTRONICS
Department of Military Affairs
idco Your power to grow Odisha Industrial Infrastructure Development Corporation
MILITARY ENGINEER SERVICES
सेल SAIL
BHEL
OFK
ओएनजीसी ONGC
WBPDCL

OVERSEAS CLIENTS & PARTNERS

SIRUI
DrukGreen
CAHORS
AL-AHLEIA SWITCHGEAR
ELITE ENGINEERING & TRADING LLC, OMAN
KERI
宝光
PVA TePla
LucasMilhaupt
Southern States
ORMAZABAL velatia
J&P SINCE 1875
Time In Staig Germany
BATEL elektromekanik
ENERGY SAVERS Technical Services LLC
DP WORLD
oman oil and gas.com

NAMES THAT TRUST US FOR FOUR DECADES

Client List of Major Distribution Companies Using BMC Make Medium Voltage CT, PT and Major Epoxy Components.

BSES

KSEB

GEB

INDIA POWER
Adding power to life

UHBVN
Uttar Haryana Bijli Vitran Nigam

RVUN

UPPCL

OPGC
Odisha Power Generation Corporation

एनटीपीसी
NTPC

DHBVN
Dakshin Haryana Bijli Vitran Nigam

TPCODL
TP CENTRAL ODISHA DISTRIBUTION LIMITED

WBSETCL

बेस्ट

PSPCL
Punjab State Power Corporation Limited

WBSEDCL

TANGEDCO

Overseas Electricity Distribution Companies

KAHRAMAA
Qatar General Electricity & Water Corporation

TENAGA NASIONAL BERHAD
MALAYSIA

هيئة كهرباء ومياه دبي
Dubai Electricity & Water Authority

PRIVATE CLIENTS

RUNGTA GROUP
Creating Infrastructure

OMPL

TOSHIBA

TDK

TRUVOLT

MYTRAH

JSW
Better Everyday

EXIDE

SUZLON
POWERING A GREENER TOMORROW

SEIL
SKIPPER

PE SYSTEM

SMARU
TECH LLP

ERG
EVOLVING ENERGIES

greenko

Shreem
ELECTRIC LTD

RASHMI
GROUP

NORTON
SAINT-GOBAIN

BOSCH

KISWOK
Engineering Better Tomorrows

POMMIER
innovative solutions for trucks

SHYAM STEEL

OMPL

SHAKAMBHARI
GROUP

MP BIRLA
GROUP

PCBL
CHEMICAL

BMC
EXCELLENCE REDEFINED
Since 1986

BMC ELECTROPLAST PVT. LTD.

Reg. Office & Works1:130/22, B.S. Industrial Estate, Hanspukur, Bakhrahat Road, Kolakta - 700104(W.B.) India
Works 2 : K-133, Waluj, Aurangabad-431136(Maharastra), India

+91 93313 32975 // +91 87775 87692
: skb@bmcepl.col, siddhartha@bmcepl.com, bhargava@bmcepl.com, samrat@bmcepl.com, saptarshi@bmcepl.com
connect@bmcepl.com

TECHNOLOGY CENTRE

Our contribution to Technology through a Technical Literature....

Available on amazon both hard and soft version...